

Sinhala-English Multilingual AI Call Center Bot with Sentiment-Aware Dialogue and Multimodal CSAT Prediction

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Abstract—This paper presents a Sri Lanka focused, voice-based call center automation framework supporting Sinhala, English, and Sinhala-English code-mixed conversations in low-resource environments. The system adopts an end-to-end architecture integrating bilingual dataset construction, privacy-preserving speech processing, retrieval-grounded response generation, multimodal sentiment intelligence, and customer satisfaction (CSAT) estimation. A Sinhala-English call-center corpus is created using noise reduction, speaker diarization, and transcript alignment, combined with transcript-aligned PII redaction. During live interaction, language identification routes calls to unified processing pipelines. Real-time sentiment analysis with explainable risk scoring supports escalation decisions, while retrieval-augmented generation ensures factually grounded responses. Emotion-adaptive text-to-speech enhances conversational naturalness. The framework enables interaction-based CSAT estimation without relying solely on post-call surveys, providing scalable and privacy-aware automation tailored to multilingual Sri Lankan call center operations.

Index Terms—Multimodal sentiment analysis, Retrieval-augmented generation, Code-mixed speech processing, Privacy-preserving speech processing, Customer satisfaction prediction

I. INTRODUCTION

Voice-based conversational AI is increasingly deployed in contact centers to improve scalability, reduce operational cost, and provide continuous customer support. Modern systems integrate Automatic Speech Recognition (ASR), Natural Language Processing (NLP), and neural text generation to automate routine interactions while maintaining a human-like conversational experience. Low-latency voice agent architectures demonstrate that streaming ASR, efficient language models,

and real-time Text-to-Speech (TTS) can be combined within unified pipelines suitable for telecommunication environments [7]. However, reliable deployment in real-world call centers remains challenging due to noisy channels, accent variability, domain-specific queries, and the need for trustworthy response generation. In addition, many organizations rely on post-call Customer Satisfaction (CSAT) surveys, which suffer from low response rates and delayed feedback, limiting their effectiveness for real-time quality monitoring.

These challenges are amplified in Sri Lanka, where contact centers frequently handle Sinhala, English, and Sinhala-English code-mixed conversations. Sinhala is a low-resource language in speech and NLP research, with early ASR corpora based on small-scale read speech [1] and later work expanding textual resources from social media [2]. However, these datasets do not capture spontaneous service-domain interactions typical of call centers. The SiTa dataset provides diarized Sinhala and Tamil speech in real-world conditions [3], but does not model structured customer-agent dialogue or downstream tasks such as sentiment estimation. Code-switching further complicates deployment, as Sinhala-English conversations involve intra-sentential switching, lexical borrowing, and non-standard forms. Although multilingual models improve cross-lingual generalization [5], they struggle with code-switched inputs when such patterns are underrepresented [6], affecting transcription reliability, sentiment interpretation, and response grounding in noisy and accented environments.

Privacy introduces another essential design constraint. Call recordings frequently contain personally identifiable information (PII) and sensitive disclosures. Traditional preprocess-

ing steps such as noise reduction and diarization improve structural organization but do not inherently address privacy risks. Research emphasizes preserving spoken content while protecting sensitive information [4], highlighting the balance between intelligibility and privacy. In bilingual call-center settings, privacy-preserving mechanisms must be integrated into the processing pipeline to enable ethical data usage without degrading linguistic and emotional information required for downstream modeling.

Beyond transcription and privacy, effective automation requires sentiment-aware reasoning and satisfaction modeling. Early ASR-driven interactive voice response systems demonstrated that in-call sentiment signals can inform dialogue management and routing decisions [15]. Multimodal sentiment analysis combining text and audio improves robustness compared to text-only approaches [16]. CSAT estimation research shows that conversational features can predict satisfaction without relying solely on surveys. Representation learning [17], speaker-turn-aware modeling [18], and sequential dialogue modeling [19] demonstrate that satisfaction signals can be inferred from interaction dynamics. However, most studies evaluate these techniques offline and do not integrate them into real-time multilingual conversational systems.

Reliable response generation presents an additional challenge. Large language models may produce fluent but ungrounded answers, introducing hallucination risk in customer support settings. Retrieval-Augmented Generation (RAG) grounds model outputs in external knowledge bases, improving factual consistency [11]. Extensions such as LA-RAG and RASU explore retrieval-enhanced speech pipelines [12], [13]. Nevertheless, integrating retrieval, multilingual processing, sentiment intelligence, and satisfaction-aware logic within a single deployable framework remains limited, particularly for low-resource, code-mixed environments.

This paper presents a Sri Lanka focused, end-to-end voice-based call-center automation framework for Sinhala, English, and Sinhala-English interactions. The system includes a bilingual dataset pipeline with speech enhancement, speaker diarization, and transcript alignment, along with transcript-aligned PII redaction for privacy compliance [4]. During live operation, incoming speech is processed through ASR and language identification, followed by sentiment inference and intent-aware routing. Sinhala interactions use prosody-based sentiment features fused with text signals, while English interactions employ a transformer-based emotion classifier. For knowledge queries, enterprise documents are retrieved and integrated into a constrained RAG framework [11], with cosine similarity used for grounding validation. Responses are generated using emotion-adaptive TTS, and all interactions are logged to support CSAT modeling [17], [19].

II. RELATED WORK

A. Sinhala–English Bilingual and Code-Mixed Dataset

Speech resources for Sinhala remain limited compared to high-resource languages. Early Sinhala Automatic Speech Recognition (ASR) corpora were primarily developed using

small-scale read speech recordings [1], while later efforts expanded text-based corpora from social media sources [2]. Although these datasets support foundational language modeling, they do not reflect spontaneous, domain-specific service interactions typical of call-center environments. The SiTa dataset provides diarized Sinhala and Tamil speech recorded under real-world conditions [3], yet it does not model structured service dialogues nor incorporate privacy-preserving processing mechanisms.

Parallel advances in speaker anonymization research have explored techniques for protecting speaker identity while preserving intelligibility and linguistic content [1]. Studies on vocoder drift and multilingual anonymization highlight the trade-offs between privacy and speech quality [2], [3], while character-level vocoder conditioning has been proposed to better preserve spoken content during anonymization [4]. However, existing anonymization research has not been explicitly tailored to bilingual, code-mixed service-domain conversations. Consequently, there remains no publicly available Sinhala-English conversational corpus designed for call-center analytics that integrates diarization, dialogue structure, and privacy-aware processing. This gap motivates the development of a structured bilingual dataset aligned with real-world service communication patterns and deployment constraints.

B. Multilingual Call Center Automation

Large-scale multilingual representation learning models, such as XLM-R, have demonstrated cross-lingual generalization through shared embeddings [5]. However, multilingual models often struggle with intra-sentential code-switching when mixed-language examples are underrepresented in training data [6]. This challenge is particularly pronounced in Sinhala-English conversational settings, where code-switching is common and context-dependent.

Recent work shows that low-latency voice agents integrating streaming ASR, large language models, and neural Text-to-Speech (TTS) are technically feasible in telecommunication environments [7]. Accent adaptation via transfer learning improves ASR performance for South Asian English accents [8], [9], while accent embeddings and multi-task learning enhance robustness to phonetic variability [10]. Retrieval Augmented Generation (RAG) has been introduced to ground language model outputs in external knowledge bases, improving factual consistency [11]. Extensions such as LA-RAG and RASU demonstrate the integration of retrieval with ASR and speech understanding pipelines [12], [13]. Sinhala conversational systems have also been explored in limited forms [14]. Nevertheless, most prior work addresses ASR, retrieval, or multilingual modeling in isolation. Comprehensive bilingual call-center frameworks that combine accent robustness, code mixed handling, and retrieval-grounded response generation remain scarce.

C. CSAT Prediction from Text

CSAT has traditionally been measured through post-interaction surveys, which often suffer from low response

rates and delayed insights. To address these limitations, early systems incorporated sentiment analysis into Interactive Voice Response (IVR) systems to adapt dialogue behavior [15], and subsequent research demonstrated that textual sentiment and conversational features correlate with satisfaction outcomes [16]. However, despite these advancements, traditional approaches still rely on indirect evaluation methods, limiting their effectiveness for real-time satisfaction assessment in conversational systems.

More recent approaches apply representation learning for automated CSAT estimation from transcripts [17]. Multi task adversarial networks incorporating speaker-turn awareness improve robustness in satisfaction prediction [18], while sequential dialogue act modeling captures conversational flow to enhance estimation accuracy [19]. However, most CSAT prediction models are evaluated offline and are not integrated within real-time conversational pipelines. Furthermore, research addressing Sinhala or code-mixed low-resource contexts remains limited.

D. Audio-Based CSAT and Prosodic Modeling

Speech Emotion Recognition (SER) research initially relied on handcrafted acoustic features such as pitch, zero-crossing rate (ZCR), RMS energy, and Mel-frequency cepstral coefficients (MFCCs) [20], [21]. With the advancement of deep learning, convolutional and recurrent neural networks were applied to spectrogram representations for improved emotion detection [22], [23]. Hybrid CNN-LSTM and attention-based models further capture temporal dynamics in conversational speech [24]. However, deep neural SER approaches may introduce computational overhead and reduced interpretability, limiting their suitability for low-resource real-time systems.

Recent work emphasizes multimodal fusion of textual and acoustic signals to improve robustness in noisy environments [16], [25]. Emotion-aware conversational TTS enables prosodic modulation conditioned on detected emotional states [26], [27]. Moreover, few studies integrate multimodal sentiment detection, retrieval-grounded response generation, and satisfaction-aware decision mechanisms within a unified bilingual conversational architecture.

Overall, existing research advances anonymization, multilingual modeling, ASR adaptation, retrieval grounding, sentiment analysis, and CSAT estimation as largely independent components. However, privacy-preserving bilingual Sinhala-English conversational corpora tailored to call-center analytics remain unavailable. Real-time integration of ASR, RAG grounding, multimodal sentiment reasoning, and satisfaction-aware decision logic in low-resource code-mixed environments is still limited, motivating the integrated framework proposed in this study.

III. METHODOLOGY

A. System Architecture Overview

The proposed system is designed as an integrated multilingual call center intelligence framework consisting of four main components: (1) prosody-preserving Sinhala-English dataset

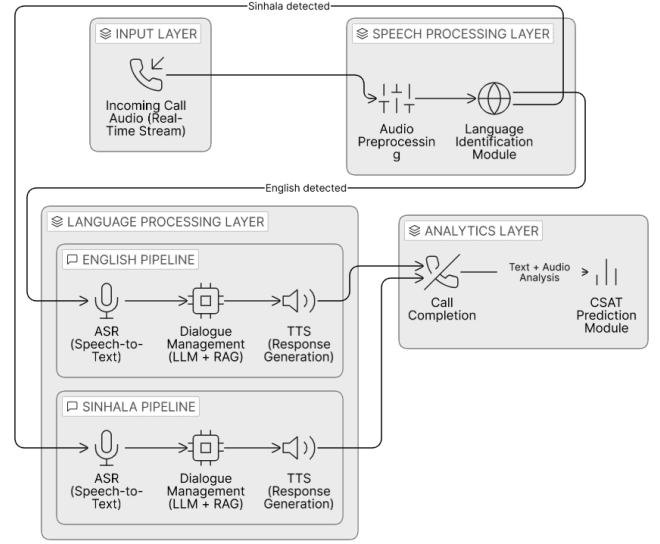


Fig. 1: Overall system architecture for real-time multilingual call handling. Incoming call audio is preprocessed and routed through language-specific pipelines (ASR → LLM+RAG → TTS), followed by CSAT prediction.

creation, (2) English call handling with CSAT prediction, (3) Sinhala call handling with text-based CSAT scoring, and (4) Sinhala audio-based CSAT prediction. As illustrated in Fig. 1, the architecture begins with incoming call audio first undergoes preprocessing, including noise reduction and signal normalization. For dataset construction, transcription and privacy-preserving anonymization are applied to produce a structured bilingual corpus. In the real-time call handling pipeline, only audio enhancement and feature extraction are performed before language-specific processing and CSAT estimation.

B. Data Collection

The dataset consists of real-world Sinhala-English call center conversations collected from a Sri Lankan service environment. The recordings capture spontaneous interactions between agents and customers, including Sinhala, English, and naturally occurring code-mixed speech. These authentic service dialogues form the foundation for building a conversational corpus that supports multilingual modeling, sentiment analysis, and CSAT prediction.

C. Sinhala-English Bilingual and Code-Mixed Dataset Construction

This component describes the development of a processing pipeline for constructing a privacy-compliant Sinhala-English call center speech corpus. The pipeline brings together audio cleaning, speaker diarization and transcript-level Personally Identifiable Information (PII) redaction in a single workflow. The main objective is to retain the natural flow and emotional characteristics of conversations while removing the information that identifies the speaker.

1) *Data Cleaning and Preparation*: All recordings are first standardized to 16 kHz mono WAV format to maintain consistency throughout the corpus. Amplitude normalization is used to scale signals to a range of $[-1, 1]$.

Given the prevalence of background noise in call center environments, spectral-domain noise reduction is used to increase clarity. Prosodic cues like pitch and speech tempo are carefully preserved because they are necessary for emotion-aware modelling. This step additionally boosts transcription quality and stabilizes feature extraction.

Each call has two participants: an agent and a customer. Speaker diarization is done with *pyannote.audio*, which uses Voice Activity Detection, Speaker Change Detection, and Agglomerative Clustering to determine speaker boundaries.

For dataset construction, diarization outputs are used to label utterances with speaker identity, enabling structured dialogue reconstruction. For anonymization, segments belonging to the same speaker are grouped so that identity transformation is applied consistently across that speaker's utterances using an Emotion-Preserving Prosody Anonymization (EPPA)-based approach. This method has been shown to effectively balance privacy and speech utility by preserving emotional characteristics while minimizing identity leakage, with only minimal impact on speech quality [28]. Recordings are then divided into 3–10 second segments for efficient processing, with shorter segments zero-padded when necessary.

2) *Transcription and PII Redaction*: A bilingual ASR system generates time-aligned transcripts, and each utterance is labeled as Sinhala, English, or code-mixed to reflect real conversational patterns in Sri Lankan service interactions.

Personally identifiable information, such as names and account numbers, is detected at transcript level using contextual and rule-based matching. Word-level timestamps allow these tokens to be aligned precisely with audio. Identified segments are masked in both text and waveform, ensuring privacy while maintaining conversational continuity.

D. English Call Handling and CSAT Prediction

The English pipeline is designed as a real-time, end-to-end conversational flow that integrates speech transcription, emotion-aware, knowledge-grounded response generation, and structured logging to support future CSAT modeling. Incoming user audio is first processed by an Automatic Speech Recognition (ASR) module to generate a textual transcript. The transcript is then analyzed for emotion and intent before downstream routing. All conversational events are logged with associated metadata to ensure traceability and enable satisfaction modeling. The architecture is implemented without reliance on paid external speech APIs, ensuring deployability in constrained environments.

1) *English Call Handling and Response Generation*: Speech transcription is performed using the Whisper large v3 model. Audio input is normalized to 16 kHz mono prior to inference. The ASR system produces a single transcript per conversational turn, which is forwarded to emotion classification and routing components.

The transcribed text is processed using a fine-tuned DistilRoBERTa-based transformer model trained for seven emotion categories: neutral, happiness, surprise, sadness, anger, disgust, and fear. Tokenization is capped at 128 tokens. To mitigate class imbalance, neutral-class downsampling was applied during training. Cross-entropy loss was used for optimization, and macro-F1 served as the primary model selection criterion.

A rule-based intent classifier categorizes each transcript into greeting, question, complaint, or feedback. Question-type queries trigger the retrieval pipeline, complaint-type inputs generate short empathetic responses, and greeting or feedback inputs are handled using predefined templates. This routing mechanism reduces unnecessary retrieval operations and improves response efficiency.

2) *Retrieval, Grounding, and Validation*: A domain-specific knowledge base was constructed from enterprise PDF documents. Text was extracted, cleaned, and segmented into overlapping chunks using a recursive character-based splitter. Each chunk was embedded using the sentence-transformer model all-MiniLM-L6-v2. Embeddings were L2-normalized and indexed using FAISS with inner-product similarity search.

For knowledge-seeking queries, the system retrieves the top-k relevant chunks based on cosine similarity. Retrieved passages are injected into a constrained Retrieval-Augmented Generation (RAG) prompt to ensure responses are grounded in verified enterprise context. The Large Language Model is configured for deterministic generation. To reduce hallucination risk, embedding-level cosine similarity between the generated answer and retrieved context is computed to produce a grounding confidence score.

3) *Emotion-Adaptive Speech Synthesis and Deployment*: The validated textual response is converted to speech using an edge-based neural Text-to-Speech engine. Prosodic parameters including pitch, speaking rate, and volume are dynamically adjusted according to the detected emotion and its confidence score.

The system is deployed using a Gradio-based web interface that supports live microphone input and streaming conversational interaction. The backend processes each conversational turn sequentially while maintaining conversation state and metadata logs for traceability and future CSAT estimation.

E. Sinhala Call Handling and Text-Based CSAT Prediction

This study proposes a real-time Sinhala–English code-mixed conversational system for call center automation, integrating Automatic Speech Recognition (ASR), Retrieval-Augmented Generation (RAG), a Large Language Model (LLM), and Customer Satisfaction (CSAT) prediction.

1) *Automatic Speech Recognition (ASR)*: Google Cloud Speech-to-Text API is used to transcribe real-time customer and agent speech into text. The API supports multilingual transcription and is configured to recognize Sinhala and English code-mixed speech. The transcribed text is passed to the downstream language understanding and retrieval modules.

2) *Retrieval-Augmented Generation (RAG)*: A RAG framework is implemented using official documentation and knowledge sources. Relevant documents are embedded and indexed using a vector database. During a conversation, the system retrieves the most relevant context passages based on the user query and provides them to the LLM to generate accurate and domain-specific responses.

3) *LLM Integration*: The OpenAI API is used as the core conversational LLM. The model receives the user query along with retrieved contextual information from the RAG system and generates natural language responses in Sinhala–English code-mixed format. Prompt engineering is applied to ensure domain-specific and polite customer service responses.

4) *CSAT Prediction*: Since an explicit CSAT module was not implemented, a sentiment-based CSAT estimation method is proposed. The transcribed conversation text is analyzed using a sentiment classification model to predict emotional polarity. A fine-grained scoring mechanism (e.g., mapping sentiment confidence scores to a 1–5 CSAT scale) is recommended to estimate customer satisfaction at the end of each call.

F. Sinhala Audio-Based CSAT Prediction

This aims to identify customer emotions in real time during ongoing call center interactions and make explainable escalation decisions. The system primarily focuses on speech-driven sentiment analysis and initially employs a prosody-based approach due to its interpretability and low computational complexity compared to deep neural network classifiers. The prosodic engine identifies prominent acoustic features from the audio stream, such as zero-crossing rate (ZCR) to identify speech sharpness, root mean square (RMS) energy to identify loudness, pitch related features, and calculated arousal values derived as a function of ZCR and energy. These features are analyzed using a sigmoid based formulation to produce a negative probability score (*neg_prob*), which identifies the probability of negative emotions. The model is conceptually calibrated against benchmark emotional speech datasets such as RAVDESS and further tuned to reflect Sinhala speech characteristics.

To enhance robustness and accuracy, additional acoustic descriptors and Mel-spectrogram representations are extracted from overlapping sliding windows. Deep learning architectures, including Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM) networks, and a hybrid CNN-LSTM model, are developed to capture both spatial and temporal emotional dynamics. A multimodal fusion engine incorporates the resulting audio-based negative probability with text sentiment outputs, and escalation decisions are based on a weighted risk score. Automated bot handling, conditional review, and real-time transfer to a human agent are made possible by threshold-based rules. In order to provide thorough end-to-end sentiment monitoring and service optimization for Sinhala contact centers, a CSAT module gathers ratings and optional Sinhala comments at the conclusion of the call.

IV. RESULTS AND DISCUSSION

The proposed system was evaluated for multilingual call handling and multimodal CSAT prediction across English and Sinhala conversations. The performance of the main components is summarized in Tables I and II.

A. English Pipeline Performance

In the English pipeline, the text-based emotion classifier achieved an accuracy of 0.8146 and a weighted F1-score of 0.8223 after applying augmentation and focal loss to address class imbalance. The lower macro-F1 score (0.4187) highlights continued difficulty in detecting rare emotional classes, although incorporating conversational context improved prediction stability across dialogue turns.

TABLE I: English Emotion Classification Results

Metric	Value
Accuracy	0.8146
Weighted F1	0.8223
Macro F1	0.4187

B. Sinhala Pipeline Performance

The Sinhala text-based models show moderate performance, with higher accuracy observed in CSAT prediction compared to sentiment classification. The audio sentiment component, trained using MFCC, pitch, and energy features with a CNN architecture, demonstrated strong sensitivity to negative sentiment, which is important for escalation in call center scenarios. Most errors occurred between neutral and mildly positive samples, reflecting subtle acoustic similarities.

TABLE II: Sinhala Text and Audio-Based Results

Component	Metric	Value
Text-based CSAT	Accuracy	0.85
Text Sentiment Analysis	Accuracy	0.60
Audio Sentiment (Negative)	F1-score	0.81
Audio Sentiment (Neutral)	F1-score	0.60
Audio Sentiment (Positive)	F1-score	0.35

For both English and Sinhala modules, CSAT is estimated using a multimodal strategy that integrates transcript-level sentiment with audio-based emotional cues. In the Sinhala pipeline, sentiment detection operates alongside a Retrieval-Augmented Generation (RAG) framework to produce contextually grounded responses while extracting satisfaction indicators. The combined use of textual and acoustic features enhances the robustness of satisfaction prediction, although performance remains influenced by class imbalance and low-resource constraints.

V. CONCLUSION

This research addressed the practical challenge of building a multilingual, privacy-aware, sentiment-aware voice automation framework for Sri Lankan call centers operating under low-resource and code-mixed conditions. The proposed system integrates Sinhala, English, and Sinhala-English call handling

within a unified real-time architecture combining ASR, language identification, sentiment inference, intent-aware routing, RAG, grounding validation, emotion-adaptive TTS, structured logging, and CSAT-oriented data capture.

A key contribution is the development of a novel dataset and preprocessing pipeline, incorporating noise reduction, diarization, transcript alignment, and transcript-aligned PII redaction. The English pipeline employs Whisper ASR, a transformer-based emotion classifier, rule-based intent routing, FAISS + MiniLM retrieval, cosine similarity grounding, and neural TTS. The Sinhala pipeline emphasizes explainable prosody-based sentiment features and fusion-based escalation scoring. Overall, the framework supports scalable automation, grounded response generation, and sentiment-aware decision-making in resource-constrained environments.

However, several limitations remain. Model performance is constrained by dataset size and limited rare-emotion coverage. Domain shift and ASR errors can propagate through the pipeline, affecting downstream accuracy. The Sinhala sentiment module prioritizes interpretability over deep modeling capacity, and large-scale benchmarking remains limited.

FUTURE WORK

Future work will focus on explicitly incorporating ASR confidence into sentiment fusion and response strategies, enabling uncertainty-aware routing and clarification policies. Broader evaluation under realistic call-center noise conditions and systematic ablation studies (text-only vs. audio-only vs. fusion) are needed. Expanding multilingual datasets with improved role separation and refined annotation guidelines will strengthen generalization. Deployment optimization through model quantization, ONNX acceleration, and latency reduction will improve scalability. Enhanced PII detection for code-mixed speech and anonymization auditing will strengthen privacy guarantees. Finally, CSAT modeling will evolve toward turn-level and longitudinal satisfaction tracking, supporting real-time trend monitoring and adaptive escalation policies. Additionally, evaluating the system against standardized benchmarks for Sinhala and code-mixed speech will be considered as an essential direction in future work.

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